The 128-Dimensional SIFT Descriptor

1 Setup: which image, and the one input you already have

A keypoint carries $(s_{\text{idx}}, y, x, \sigma)$ — one row of the $(N, 4)$ tensor. The descriptor is built from the **spatial gradients of a single Gaussian-blurred image**: the slice chosen by matching the keypoint's σ against the octave's σ ladder. *Not* the DoG, and *not* the 3-D scale volume.

Once that slice is picked, **everything below is 2-D image space**. Scale has only two remaining jobs: it selected the slice, and (through σ) it sets the size of the sampling window.

2 Step 1 — Gradients as magnitude and angle

On the chosen blurred image L, at each pixel, from the two axis-aligned central differences:

$$G_x = L_{x+1} - L_{x-1}, \qquad G_y = L_{y+1} - L_{y-1},$$
$$m = \sqrt{G_x^2 + G_y^2}, \qquad \theta = \operatorname{atan2}(G_y,\ G_x).$$

The magnitude m is edge strength; the angle θ is edge direction. The angle is *measured* from the two differences — never requested as an input.

3 Step 2 — Dominant orientation (rotation invariance)

Before building anything, find the keypoint's own "up":

- Over a region around the keypoint (radius $\propto \sigma$), build a **36-bin** histogram (10° each) of θ.
- Each pixel votes by its magnitude m, Gaussian-weighted so nearer pixels count more.
- The **peak bin** is the keypoint's reference orientation.

Every angle used afterward is measured *relative* to this reference:

$$\theta_{\text{relative}} = \theta_{\text{measured}} - \theta_{\text{dominant}}.$$

> Rotation invariance comes entirely from this subtraction: rotate the frame to the feature's own dominant direction, so the same feature photographed rotated yields the same descriptor. The image is still only ever differenced along x and y — the rotation is arithmetic on the angle, not a tilted derivative.

If a second histogram peak reaches $\geq 80\%$ of the maximum, a **second keypoint** is emitted at the same (x, y, σ) with that orientation — so one detection can produce two descriptors (descriptor count $\geq$ keypoint count).

4 Step 3 — A 4×4 grid of cells

Center a window on the keypoint, **rotated** to the dominant orientation and sized $\propto \sigma$, and split it into a 4×4 arrangement of subregions:

$$4 \times 4 = 16 \text{ cells.}$$

Because the grid is rotated, its sample points fall at *fractional* image locations, so L is read there by **bilinear interpolation**.

5 Step 4 — An 8-bin orientation histogram per cell

Within each of the 16 cells, build an **8-bin** histogram ($45°$ each) of the gradient orientations there — each pixel voting by m, Gaussian-weighted from the center, all angles *relative to the dominant orientation.*

6 Step 5 — That is the 128

$$\underbrace{4 \times 4}_{\text{spatial cells}} \times \underbrace{8}_{\text{orientation bins}} = 128.$$

Flatten the 16 per-cell histograms into one vector of length 128.

7 Step 6 — Normalize (illumination invariance)

1. Normalize the 128-vector to unit length — cancels contrast / gain (multiplicative light changes).
2. Clamp every element to ≤ 0.2 — stops one huge gradient from dominating (robustness to non-linear illumination, e.g. glare).
3. Renormalize to unit length.

The result is the final descriptor.

8 The interpolation that separates working SIFT from fragile SIFT

Three places, all so the descriptor changes *smoothly* as the keypoint shifts slightly — essential for matching across viewpoints:

- **Rotated window** $\rightarrow$ bilinear image interpolation at fractional sample points (Step 3).
- **Trilinear vote spreading:** each gradient sample is split across neighbouring cells (2 spatial axes) and neighbouring orientation bins (1 angular axis), weighted by distance to each bin center, so a pixel near a boundary does not jump discontinuously between cells. *This is the single biggest contributor to descriptor stability.*
- σ **scales the window** (Step 3) — the only place scale re-enters, just a multiplier on the sampling radius.

9 Why this design gives the invariances

Property	**Where it comes from**
Scale invariance	descriptor built in the keypoint's octave, window sized by σ
Rotation invariance	all angles relative to the dominant orientation (Step 2)
Illumination invariance	unit-normalize $\rightarrow$ clamp 0.2 $\rightarrow$ renormalize (Step 6)
Small-shift robustness	Gaussian weighting + trilinear vote spreading (Step 4)

That combination — scale-, rotation-, and light-invariant, and stable under small position error — is what lets the *same* descriptor match across two photos taken from different distances and angles: the whole reason the descriptors are computed for reconstruction.